Fiscal Year:	FY 2020	**Task Last Updated:**	FY 10/07/2020
PI Name:	Asadi, Ebrahim Ph.D.		
Project Title:	Enhancement and Verification of Quantitative Phase-Field Crystal Modeling using NASA-PSI Coarsening in Solid-Liquid Mixtures Experiments Data		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	38152-0001	**Congressional District:**	9
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
Start Date:	01/01/2017	**End Date:**	12/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2019 per NSSC (Ed., 10/5/20) NOTE: Change in period of performance to 1/01/2017-12/31/2018 (originally 9/19/2016-9/18/2018) per PI and NSSC information (Ed., 9/20/17)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Laradji, Mohamed Ph.D. (University of Memphis)		
Grant/Contract No.:	NNX16AT77G		
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Task Description:	Verification and enhancement of predictive computational material models for solid-liquid mixtures based on experiments conducted on Earth is challenging because of the detrimental gravitational effects on solid particles and liquid mixtures, e.g., buoyancy-induced convection and particle sedimentation. Microgravity environment for NASA-PSI (Physical Sciences Informatics) “coarsening in solid-liquid mixtures experiments” provides data without the detrimental gravitational effects that can be used for verification and enhancement of predictive models. In return, the verified/enhanced model can be used to predict coarsening in solid-liquid mixtures that ideally eliminates the need to conduct similar experiments for other materials and also experiments for other phenomena involved with solid-liquid coexisting. The mentioned predictive capability is only possible if the computational model is not derived phenomenologically for the specific application and material; phase-field crystals (PFC) are a class of such predictive computational models. Therefore, the objectives of this proposal are: 1) The development of a quantitative multi-time and length-scale computational modeling framework and tool around PFC to predict coarsening in Pb-Sn solid-liquid mixtures. Challenges for this objective are quantifying the PFC model and developing an efficient computational tool for three-dimensional (3D) simulations. 2) Data mining of NASA-PSI Coarsening in Solid-Liquid Mixtures (CSLM), CSLM-2, CSLM-2R, and CSLM-3 experiments and using them in series of simulations in the developed computational tool for enhancement and verification of the proposed modeling framework. The challenge for this objective is the huge number of 3D simulations that require significant amount of computer power. 3) Quantifying the developed model and tool to Al-Cu solid-liquid mixtures to show the potential benefits of the proposed research. The challenge for this objective is only the quantification process of the model.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed computational tool verified/quantified based on CSLM data will establish a virtual laboratory that can be used to study coarsening as well as other material processing phenomena on Earth involving with solid-liquid coexistence; e.g., rapid solidification and grain growth. Since computational models are highly repeatable at lower cost, the proposed modeling framework will result in accelerated material process development and certification. Since solid-liquid coexistence is the initial step in many manufacturing technologies such as casting, liquid infiltration, and additive manufacturing, this modeling framework can be used to investigate the relations between these manufacturing processes and the material properties in order to optimize/control the manufacturing processes parameters. In addition, the results of this research will be disseminated to the public in forms of peer-reviewed articles, scientific reports, a project-specific website, and data uploaded to NASA-PSI website.
Task Progress:	The modified embedded-atom method (MEAM) binary parameters have been fitted to the lattice parameter of Pb-rich fcc at 483K and ß-Sn-rich solution at 298K. In addition, enthalpy of mixing of solid and liquid Pb-Sn has been included in the fitting process. The results show that the present binary Pb-Sn MEAM potential can successfully reproduce the enthalpy of mixing in both solid and liquid phases over a significant range of Sn concentrations. The verification of binary Pb-Sn potential has been carried out by calculating the solidus and liquidus lines of the phase diagram for Pb-Sn system in the Sn-rich region by hybrid Molecular Dynamics/Monte Carlo simulation. The calculated phase diagram is in good agreement with available experimental data for the solidus and liquidus lines in the Sn-rich region of the phase diagram. Due to the importance of solid-liquid interfacial free energy (?), this quantity has been calculated based on the new developed potential. We used our previous calculation of phase diagram and constructed the equilibrium solid-liquid at four temperatures, namely 463K, 473K, 500K, and 511K, based on the solidus and liquidus. Then, capillary fluctuation method (CFM) has been employed for the calculation of solid-liquid stiffness at the interface. The results shows that the orientation of solid crystal does not have influence on the value of stiffness. The calculated solid-liquid interfacial energy for pure is in good agreement with experimental counterpart. Additionally, we found that ? slightly increases with adding Pb in liquid Pb-Sn, which is in agreement with the available experiment. In addition, we simulated the kinetics of coarsening in solid-liquid mixtures (Ostwald ripening of solid domains in the liquid phase via 1PFC model. The simulations, which were performed systematically as a function of volume fraction of the solid phase, showed that dynamical scaling is reached during late times, and the growth law is in good agreement with the classical theory of Lifshitz, Slyozov, and Wagner (LSW), i.e., R~ t1/3, an indication that domain growth is mediated by the long-range inter-domain diffusion of atoms. In contrast to LSW’s theory, however, the domain size distribution is symmetric, and can be fitted with a Gaussian. The investigation of the topological domain structure, through a Voronoi tessellation of the domains’ centers of mass showed that both Lewis’ law and Aboav-Weaire law of two-dimensional cellular patterns are satisfied, implying that the kinetics proceed such as the conformational entropy of the domains-containing Voronoi cells is maximized. These results are in very good agreement with an earlier experimental study of a phase-separating phospholipid-cholesterol Langmuir film. We also uploaded the developed interatomic potential files in this project to National Institute of Standards and Technology's (NIST’s) Interatomic Potential Repository. This allows users to readily access our new potentials. The link is: https://
Bibliography Type:	Description: (Last Updated: 07/29/2025)
Articles in Peer-reviewed Journals	Moats KA, Asadi E, Laradji M. "Phase field crystal simulations of the kinetics of Ostwald ripening in two dimensions." Phys Rev E. 2019 Jan;99(1-1):012803. https://doi.org/10.1103/PhysRevE.99.012803 ; PMID: 30780278 , Jan-2019
Articles in Peer-reviewed Journals	Etesami SA, Laradji M, Asadi E. "Reliability of molecular dynamics interatomic potentials for modeling of titanium in additive manufacturing processes." Computational Materials Science. 2020 Nov; 184:109883. https://doi.org/10.1016/j.commatsci.2020.109883 , Nov-2020
Articles in Peer-reviewed Journals	Etesami SA, Laradji M, Asadi E. "The influence of Pb content on the interfacial free energy of solid Sn in eutectic Pb–Sn liquid mixtures using molecular dynamics simulations." Mol Simul. 2022 Feb 5;48(7):568-573. https://doi.org/10.1080/08927022.2022.2032693 , Feb-2022

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